

Koustubh Phalak

✉ koustubhphalak.kp@gmail.com

🌐 [Linkedin](#)

🌐 [Personal Homepage](#)

☎ +1(814)-996-8980

🎓 [Google Scholar](#)

Research Interests

Quantum Computing, Quantum Algorithms, Machine Learning, Generative AI

Work Experience

Aug 24 - now

📌 **Graduate Teaching Assistant, Penn State**

- Working as Teaching Assistant for introductory Python course **CMPSC 131 - Programming and Computation I: Fundamentals**
- Duties include **conducting recitation sections** for timed assignments and aid students with whatever doubts they have, grading them and **help in making and grading quizzes**

Jun – Aug, 2024

📌 **Quantum Algorithms Scientist, Rigetti Computing**

- Worked on implementing **novel qubit efficient QAOA algorithm** for large scale combinatorial optimization problems
- Built **multi-qubit gates for QAOA circuit from scratch** while taking hardware constraints into account
- Ran qubit efficient QAOA circuits on **Rigetti Ankka-9Q-3 QPU**.

Aug 21 – May 24

📌 **Graduate Research Assistant, Penn State**

- Worked on my PhD research to publish works in multiple journals and conferences
- Mentored junior PhD students to get them started with their research

Education

2020 –

📌 **Ph.D. Computer Science, Penn State University** in Quantum Machine Learning, Quantum Security.

2016 – 2020

📌 **Undergrad., Electrical & Electronics Engineering, BITS Pilani Hyd Campus**

Research Publications (Citations: 286, h-index: 7)

Journal Articles

- 1 C. Beaudoin, **K. Phalak**, and S. Ghosh, "Predicting side effect of drug molecules using recurrent neural networks," *IEEE transactions on emerging topics in computational intelligence*, vol. 9, no. 2, pp. 2073–2078, 2024, Impact Factor: 6.5.
- 2 **K. Phalak**, A. Chatterjee, and S. Ghosh, "Quantum random access memory for dummies," *Sensors*, vol. 23, no. 17, p. 7462, 2023, Impact Factor: 3.5.

- 3 **K. Phalak** and S. Ghosh, "Shot optimization in quantum machine learning architectures to accelerate training," *IEEE Access*, vol. 11, pp. 41 514–41 523, 2023, Impact Factor: 3.6.
- 4 **K. Phalak**, J. Li, and S. Ghosh, "Trainable pqc-based qram for quantum storage," *IEEE Access*, vol. 11, pp. 51 892–51 899, 2023, Impact Factor: 3.6.
- 5 **K. Phalak**, A. Ash-Saki, M. Alam, R. O. Topaloglu, and S. Ghosh, "Quantum puf for security and trust in quantum computing," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 11, no. 2, pp. 333–342, 2021, Impact Factor: 3.8.

Conference Proceedings

- 1 **K. Phalak**, J. Li, and S. Ghosh, "Dataset distillation for quantum neural networks," in *IEEE Computer Society Annual Symposium on VLSI*, 2025, Acceptance Rate: 25–45%.
- 2 S. Upadhyay, **K. Phalak**, J. Lee, K. M. Hill, and S. Ghosh, "Quantum computing and cybersecurity education: A novel curriculum for enhancing graduate stem learning," in *American Society for Engineering Education*, 2025, Acceptance Rate: 35–35%.
- 3 **K. Phalak**, A. Ghosh, and S. Ghosh, "Optimizing quantum embedding using genetic algorithm for qml applications," in *2025 26th International Symposium on Quality Electronic Design (ISQED)*, IEEE, 2025, 1–9, Acceptance Rate: 25–35%.
- 4 **K. Phalak** and S. Ghosh, "Qualiti: Quantum machine learning hardware selection for inferencing with top-tier performance," in *2025 38th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems (VLSID)*, IEEE, 2025, 296–301, Acceptance Rate: 25–30%.
- 5 C. Beaudoin, **K. Phalak**, and S. Ghosh, "Altgraph: Redesigning quantum circuits using generative graph models for efficient optimization," in *Proceedings of the Great Lakes Symposium on VLSI 2024*, 2024, 44–49, Acceptance Rate: 24–28%.
- 6 **K. Phalak** and S. Ghosh, "Non-parametric greedy optimization of parametric quantum circuits," in *2024 25th International Symposium on Quality Electronic Design (ISQED)*, IEEE, 2024, 1–7, Acceptance Rate: 25–35%.
- 7 D. Bhattacharjee, A. Saha, J. Li, *et al.*, "Software flow for quantum computing," in *International Conference on Computational Technologies and Electronics*, Springer, 2023, 206–227, Acceptance Rate: 30–40%.
- 8 A. Chatterjee, **K. Phalak**, and S. Ghosh, "Quantum error correction for dummies," in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, IEEE, vol. 1, 2023, 70–81, Acceptance Rate: 18–25%.
- 9 **K. Phalak**, M. Alam, A. Ash-Saki, R. O. Topaloglu, and S. Ghosh, "Optimization of quantum read-only memory circuits," in *40th IEEE International Conference on Computer Design, ICCD 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, 117–123, Acceptance Rate: 18–23%.
- 10 A. A. Saki, M. Alam, **K. Phalak**, A. Suresh, R. O. Topaloglu, and S. Ghosh, "A survey and tutorial on security and resilience of quantum computing," in *2021 IEEE European Test Symposium (ETS)*, IEEE, 2021, 1–10, Acceptance Rate: 25–35%.

Preprints

- 1 **K. Phalak**, J. Li, and S. Ghosh, "Approximate quantum random access memory architectures. arxiv 2022," arXiv:2210.14804, 2022.

Key Research Projects

Optimal Quantum Embedding Search Problem

GENETIC ALGORITHM , QML

ISQED 24 

- Developed a **novel genetic algorithm** for searching optimal state preparation circuit for **Quantum Neural Networks**
- **Finds optimal angle embedding permutation** that yields best inferencing performance
- **Competitive results** with current State-of-the-Art methods

Greedy Parametric Quantum Circuit optimization

GREEDY ALGORITHM , QML

ISQED 23 

- Proposed a **novel greedy algorithm** to **optimize the post-transpilation circuit depth and gate count** of Parametric Quantum Circuit of a QNN
- Used Hilbert-Schmidt distance as main metric to **convert parametric 1Q rotation gates to equivalent non-parametric representations**
- Observed upto **14% and 48% reduction** in circuit depth and gate count respectively

Quantum Random Access Memory for Storage

QUANTUM STORAGE , CLASSIFICATION

IEEE Access 23 

- Designed a **novel trainable Parametric Quantum Circuit-based Quantum Random Access Memory** to store classical data
- Stored two types of data: **image data and binary data**
- QRAM Storage yielded **faster convergence for image data** for classification task and **error-free storage for upto 4-bit binary data**
- Additional **results from IBM Oslo and IBM Jakarta hardware** for image data classification task shows **competitive results** with state-of-the-art methods

Shot Optimization for QML Algorithms

QUANTUM CIRCUIT EXECUTIONS , QML



- Proposed a **gradient-free methodology** to **reduce total shot usage** in QML algorithms
- Presented **two methods** where the number of shots are function of the iterations completed, **one linear and one step function**
- Incorporated the methodology for **classification task** and **molecular property prediction task**, where **linear works better for first and step function performs better for later**

Research Service to Community

Reviewer



Quantum Information Processing, Quantum Machine Intelligence, The Journal of Supercomputing, Optical and Quantum Electronics, Scientific Reports

Sub-reviewer



IEEE QCE 2024, IEEE QCE 2023

Technical Skills

Programming–

Languages

Python, MySQL, MATLAB, $\text{\LaTeX}$, C++ (beginner), Unix Shell

Machine Learning/ –

Deep Learning Tools

PyTorch, Numpy, Pandas, SciPy, Matplotlib

Quantum Tools

Qiskit, PennyLane

Miscellaneous Experience

- 2025
- Alumni Award nomination, Nominated for BITS 30 under 30 alumni awards (ongoing)
 - Podcast talk, Invited to Physics Frontiers Podcast to give a talk on QRAM: [link](#)
 - Quantum Hackathon, Participated in BlueQubit Quantum Hackathon and finished at rank 17